

ADVANCED MATHEMATICAL PHYSICS

PHYS 721

11.7.1 Determine the nature of the singularities of each of the following functions and evaluate the residues ($a > 0$).

- (a) $\frac{1}{z^2 + a^2}$ (b) $\frac{1}{(z^2 + a^2)^2}$
 (c) $\frac{z^2}{(z^2 + a^2)^2}$ (d) $\frac{\sin 1/z}{z^2 + a^2}$
 (e) $\frac{ze^{iz}}{z^2 + a^2}$ (f) $\frac{ze^{iz}}{z^2 - a^2}$
 (g) $\frac{e^{iz}}{z^2 - a^2}$ (h) $\frac{z^{-k}}{z+1}$, $0 < k < 1$.

Recall:

For a singularity z_0 of n^{th} order,

$$\text{Res}_{z=z_0} [f(z)] = \frac{1}{(n-1)!} \lim_{z \rightarrow z_0} \left[\frac{d^{n-1}}{dz^{n-1}} (z-z_0)^n f(z) \right]$$

Let's find the singularities at finite z values. Then, investigate the $z \rightarrow \infty$ ones.

a) $\frac{1}{z^2 + a^2} = \frac{1}{(z+ia)(z-ia)}$ $\text{Res}_{z=ia} \left[\frac{1}{z^2 + a^2} \right] = \left. \frac{z-ia}{z^2 + a^2} \right|_{z=ia} = \left. \frac{1}{z+ia} \right|_{z=ia} = -\frac{1}{2a} i$

Singularity at $z = \pm ia$
(simple poles)

$\text{Res}_{z=-ia} \left[\frac{1}{z^2 + a^2} \right] = \left. \frac{z+ia}{z^2 + a^2} \right|_{z=-ia} = \left. \frac{1}{z-ia} \right|_{z=-ia} = \frac{1}{2a} i$

b) $\frac{1}{(z^2 + a^2)^2} = \frac{1}{(z+ia)^2(z-ia)^2}$ $\text{Res}_{z=ia} \left[\frac{1}{(z^2 + a^2)^2} \right] = \lim_{z \rightarrow ia} \left[\frac{d}{dz} \frac{(z-ia)^2}{(z^2 + a^2)^2} \right] = \lim_{z \rightarrow ia} \frac{d}{dz} \frac{1}{(z+ia)^2} = -2(z+ia)^{-3} \Big|_{z=ia} = -\frac{1}{4a^3} i$

Singularity at $z = \pm ia$
(poles of 2nd order)

$\text{Res}_{z=-ia} \left[\frac{1}{(z^2 + a^2)^2} \right] = \lim_{z \rightarrow -ia} \left[\frac{d}{dz} \frac{(z+ia)^2}{(z^2 + a^2)^2} \right] = \lim_{z \rightarrow -ia} \frac{d}{dz} \frac{1}{(z-ia)^2} = -2(z-ia)^{-3} \Big|_{z=-ia} = \frac{1}{4a^3} i$

c) Singularity behavior same as b. Singularity at $z = \pm ia$
(poles of 2nd order)

$\text{Res}_{z=ia} \left[\frac{z^2}{(z^2 + a^2)^2} \right] = \lim_{z \rightarrow ia} \left[\frac{d}{dz} \frac{z^2(z-ia)^2}{(z^2 + a^2)^2} \right] = \lim_{z \rightarrow ia} \frac{d}{dz} \frac{z^2}{(z+ia)^2} = \frac{2ia \left(1 + \frac{ia}{2}\right)^{-3}}{z^2} \Big|_{z=ia} = -\frac{1}{4a} i$

$\text{Res}_{z=-ia} \left[\frac{z^2}{(z^2 + a^2)^2} \right] = \lim_{z \rightarrow -ia} \left[\frac{d}{dz} \frac{z^2(z+ia)^2}{(z^2 + a^2)^2} \right] = \lim_{z \rightarrow -ia} \frac{d}{dz} \frac{z^2}{(z-ia)^2} = \frac{-2ia \left(1 - \frac{ia}{2}\right)^{-3}}{z^2} \Big|_{z=-ia} = \frac{1}{4a} i$

$$d) \frac{\sin(1/z)}{(z^2 + a^2)}$$

$$\text{Res}_{z=ia} \left[\frac{\sin(1/z)}{z^2 + a^2} \right] = \left. \frac{\sin(1/z)}{z + ia} \right|_{z=ia} = \frac{\sin(-1/a)}{2ia} = -\frac{\sinh(1/a)}{2a}$$

↳ Singularities:

$z=0$ essential

$z=\pm ia$ simple pole

$$\text{Res}_{z=-ia} \left[\frac{\sin(1/z)}{z^2 + a^2} \right] = \left. \frac{\sin(1/z)}{z - ia} \right|_{z=-ia} = \frac{\sin(1/a)}{-2ia} = -\frac{\sinh(1/a)}{2a}$$

$$\text{Res}_{z=0} \left[\frac{\sin(1/z)}{z^2 + a^2} \right] \Rightarrow \left(\frac{1}{z} - \frac{1}{3!z^3} + \dots \right) \left[\frac{1}{a^2} \left(\frac{1}{1 + z^2/a^2} \right) \right]$$

$$\left(\frac{1}{z} - \frac{1}{3!z^3} + \dots \right) \left[\frac{1}{a^2} \left(1 - \frac{z^2}{a^2} - \frac{z^4}{a^4} - \frac{z^6}{a^6} - \dots \right) \right]$$

$$a_{-1} = \frac{1}{a^2} \left[1 + \frac{1}{3!a^2} + \frac{1}{5!a^4} + \frac{1}{7!a^6} + \dots \right]$$

$$= \frac{1}{a} \left[\frac{1}{a} + \frac{1}{3!a^3} + \frac{1}{5!a^5} + \frac{1}{7!a^7} + \dots \right] = \frac{1}{a} \sinh(1/a)$$

$$e) \frac{ze^{iz}}{z^2 + a^2}$$

$$\text{Res}_{z=ia} [f(z)] = \left. \frac{ze^{iz}}{z + ia} \right|_{z=ia} = \frac{ia e^{-a}}{2ia} = \frac{e^{-a}}{2}$$

Singularity at $z=\pm ia$ (simple poles)

$$\text{Res}_{z=-ia} [f(z)] = \left. \frac{ze^{iz}}{z - ia} \right|_{z=-ia} = \frac{-ia e^a}{-2ia} = \frac{e^a}{2}$$

$$f) \frac{ze^{iz}}{z^2 - a^2}$$

$$\text{Res}_{z=a} [f(z)] = \left. \frac{ze^{iz}}{z + a} \right|_{z=a} = \frac{a e^{ia}}{2a} = \frac{e^{ia}}{2}$$

Singularity at $z=\pm a$ (simple poles)

$$\text{Res}_{z=-a} [f(z)] = \left. \frac{ze^{iz}}{z - a} \right|_{z=-a} = \frac{-a e^{-ia}}{-2a} = \frac{e^{-ia}}{2}$$

$$g) \frac{e^{iz}}{z^2 - a^2}$$

$$\text{Res}_{z=a} [f(z)] = \left. \frac{e^{iz}}{z + a} \right|_{z=a} = \frac{e^{ia}}{2a} = \frac{e^{ia}}{2a}$$

Singularity at $z=\pm a$ (simple poles)

$$\text{Res}_{z=-a} [f(z)] = \left. \frac{e^{iz}}{z - a} \right|_{z=-a} = \frac{-e^{-ia}}{-2a} = \frac{e^{-ia}}{2a}$$

$$h) \frac{z^{-k}}{z+1} \quad 0 < k < 1$$

We find that there is a :

* Singularity at $z=-1$ (simple pole)

$$\text{Res}_{z=-1} \left[\frac{z^{-k}}{z+1} \right] = \lim_{z \rightarrow -1} z^{-k} = (-1)^{-k}$$

* Branch point at $z=0$ → it doesn't make sense to talk about a residue for a BP.

Investigating singularities at $z \rightarrow \infty$, we are trying to find functions that blow up at $z \rightarrow \infty$. These are the ones with the term e^{iz} or parts e, f, g.

We conduct the change of variable $z = 1/w$ and investigate singularities of $-\frac{g(w)}{w^2}$

$$\oint_c f(z) dz = \oint_c g(w) \cdot \left(-\frac{dw}{w^2}\right)$$

Note all singularities at $z \rightarrow \infty$ of parts e, f, g are essential due to the $e^{i/w}$ term.

$$\begin{aligned} \text{e)} \quad -\frac{1}{w^2} \frac{1/w e^{i/w}}{1/w^2 + a^2} &= -\frac{1/w e^{i/w}}{1 + a^2 w^2} = -1/w \left(1 + \frac{i}{w} + \frac{-1}{2! w^2} + \frac{i^3}{3! w^3} + \dots \right) \left(1 - a^2 w^2 + a^4 w^4 - a^6 w^6 + \dots \right) \\ a_{-1} &= -\left[1 + \frac{a^2}{2!} + \frac{a^4}{4!} + \dots - \frac{a^{2n}}{n!} \right] = -\cosh a \end{aligned}$$

$$\begin{aligned} \text{f)} \quad -\frac{1}{w^2} \frac{1/w e^{i/w}}{1/w^2 - a^2} &= -\frac{1/w e^{i/w}}{1 - a^2 w^2} = -1/w \left(1 + \frac{i}{w} + \frac{-1}{2! w^2} + \frac{i^3}{3! w^3} + \dots \right) \left(1 + a^2 w^2 + a^4 w^4 + a^6 w^6 + \dots \right) \\ a_{-1} &= -\left[1 - \frac{a^2}{2!} + \frac{a^4}{4!} - \dots (-1)^n \frac{a^{2n}}{n!} \right] = -\cos a \end{aligned}$$

$$\text{g)} \quad -\frac{1}{w^2} \frac{e^{i/w}}{1/w^2 - a^2} = -\frac{e^{i/w}}{1 - a^2 w^2} = \left(1 + \frac{i}{w} + \frac{-1}{2! w^2} + \frac{i^3}{3! w^3} + \dots \right) \left(1 + a^2 w^2 + a^4 w^4 + a^6 w^6 + \dots \right)$$

$$\begin{aligned} a_{-1} &= -\left(i - \frac{i a^2}{3!} + \frac{i a^4}{5!} - \dots i (-1)^n \frac{a^{2n}}{(2n+1)!} \right) \\ &= -\frac{i}{a} \sum (-1)^n \frac{a^{2n+1}}{(2n+1)!} = -\frac{i}{a} \sin a \end{aligned}$$

11.8.1 Generalizing Example 11.8.1, show that

$$\int_0^{2\pi} \frac{d\theta}{a \pm b \cos \theta} = \int_0^{2\pi} \frac{d\theta}{a \pm b \sin \theta} = \frac{2\pi}{(a^2 - b^2)^{1/2}}, \quad \text{for } a > |b|.$$

What happens if $|b| > |a|$?

In an effort to solve this problem, we pursue a change in variables from $\theta \rightarrow z$.

$$\begin{aligned} z &= e^{i\theta} \\ dz &= ie^{i\theta} d\theta \\ d\theta &= \frac{dz}{iz} \end{aligned} \quad \begin{aligned} \cos \theta &= \frac{e^{i\theta} + e^{-i\theta}}{2} \\ &= \frac{z + z^{-1}}{2} \end{aligned} \quad \begin{aligned} \sin \theta &= \frac{e^{i\theta} - e^{-i\theta}}{2i} \\ &= \frac{z - z^{-1}}{2i} \end{aligned}$$

We are working with 4 cases & 2 for $\pm b$ & 2 for $\cos \theta$ or $\sin \theta$.
We solve for the positive b and $\cos \theta$ then adapt on the proof for the other cases.

$$\begin{aligned} &\int_0^{2\pi} \frac{1}{a + b \cos \theta} d\theta \\ &= \oint_C \frac{1}{a + b \left(\frac{z + z^{-1}}{2} \right)} \frac{1}{iz} dz \\ &= \frac{2}{ib} \oint_C \frac{1}{z^2 + \frac{2a}{b}z + 1} dz \\ &= \frac{2}{ib} \cdot 2\pi i \operatorname{Res}_{z \rightarrow z_0} \left[\frac{1}{(z - z_0)(z - z_1)} \right] \\ &= \frac{4\pi}{b} \cdot \frac{1}{z_0 - z_1} \\ &= \frac{4\pi}{b} \cdot \frac{1}{2\sqrt{\frac{a^2}{b^2} - 1}} = \frac{2\pi}{(a^2 - b^2)^{1/2}} \end{aligned}$$

z_0, z_1 are roots of $z^2 + \frac{2a}{b}z + 1$.

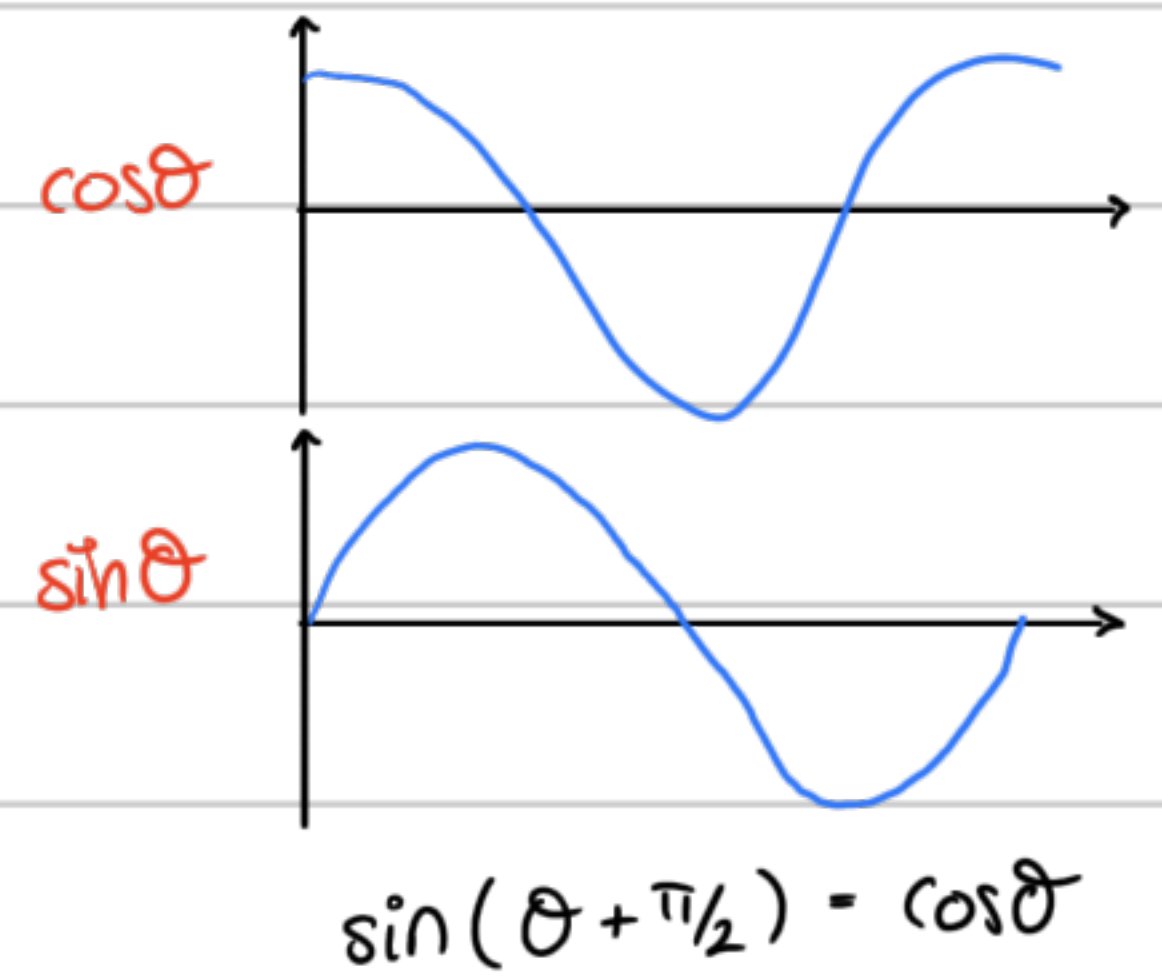
Since $z_0 z_1 = c = 1$, WLOG let $|z_0| < 1$ and $|z_1| > 1$

$$\begin{aligned} z_i &= \frac{-\frac{2a}{b} \pm \sqrt{\frac{4a^2}{b^2} - 4}}{2} & \text{since } a > b, \\ &= -\frac{a}{b} \pm \sqrt{\frac{a^2}{b^2} - 1} \\ z_0 &= -\frac{a}{b} + \sqrt{\frac{a^2}{b^2} - 1} \\ z_1 &= -\frac{a}{b} - \sqrt{\frac{a^2}{b^2} - 1} \end{aligned}$$

* For the $b < 0$ case, the result doesn't change as we have a b^2 term in the answer.

* For the $\sin \theta$ case, we can convert the $\cos \theta$ to $\sin \theta$ by a change in phase $\frac{\pi}{2}$

$$\begin{aligned} \int_0^{2\pi} \frac{1}{a \pm b \cos \theta} d\theta &= \int_0^{2\pi} \frac{1}{a \pm b \sin(\theta + \pi/2)} d\theta \\ &= \int_{\pi/2}^{5\pi/2} \frac{1}{a \pm b \sin \theta'} d\theta' \\ &= \int_0^{2\pi} \frac{1}{a \pm b \sin \theta} d\theta \quad \text{since sine is periodic \& the range is } 2\pi \\ &= \frac{2\pi}{\sqrt{a^2 - b^2}} \end{aligned}$$



If $|b| > |a|$, then $|\frac{a}{b}| < 1$ such that the denominators

$$a \pm b \cos \theta$$

$$a \pm b \sin \theta$$

have roots θ_n causing a singularity that make the integrals diverge.

11.8.9 Prove that $\int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2} dx = \frac{\pi}{2}$.

Hint. $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$.

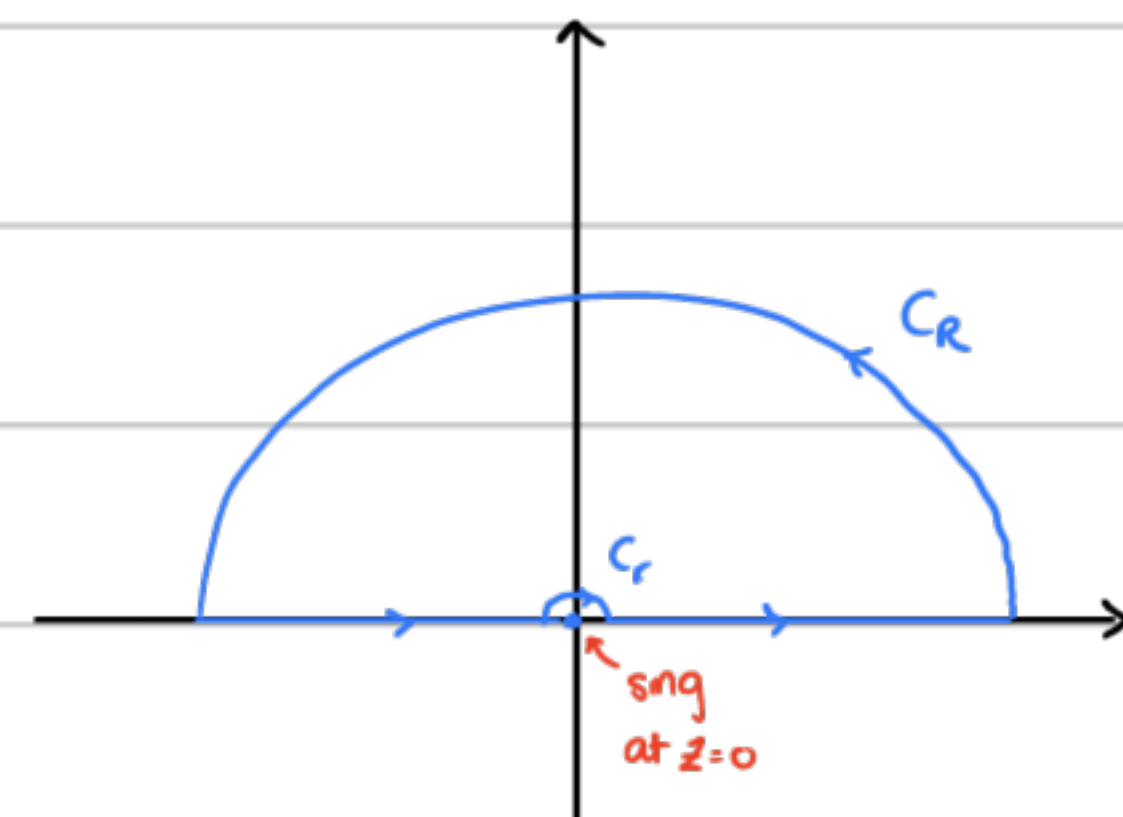
Goal is Find $\int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2} dx$

Solution

We proceed by drawing the contour on the avoiding the singularity at $z=0$.

Since the singularity is in the interval of integration, we will actually be finding the Cauchy Principal value

$$\int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2} dx$$



We rewrite the integral to make the problem more tractable &

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2} dx &= \int_{-\infty}^{\infty} \frac{\frac{1}{2}(1 - \cos 2x)}{x^2} dx \\ &= \underbrace{\frac{1}{2} \int_{-\infty}^{\infty} \frac{1}{x^2} dx}_{I_1} - \underbrace{\frac{1}{2} \int_{-\infty}^{\infty} \frac{\cos 2x}{x^2} dx}_{I_2} \end{aligned}$$

Evaluating I_1 &

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{1}{x^2} dx &= \lim_{r \rightarrow 0} \int_{-\infty}^{-r} \frac{1}{x^2} dx + \int_r^{\infty} \frac{1}{x^2} dx \\ &= \lim_{r \rightarrow 0} \int_{-\infty}^{-r} \frac{1}{x^2} dx - \int_{-\infty}^{-r} \frac{1}{x^2} dx \quad \text{change of variable } x \rightarrow -x \\ &= 0 \end{aligned}$$

Evaluating I_2 &

$$\int_{-\infty}^{\infty} \frac{\cos 2x}{x^2} dx = \operatorname{Re} \left\{ \int_{-\infty}^{\infty} \frac{e^{i2x}}{x^2} dx \right\}$$

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{e^{i2x}}{x^2} dx &= \underbrace{\int_C \frac{e^{i2z}}{z^2} dz}_{\substack{0 \\ \text{no residues} \\ \text{in } C}} - \underbrace{\int_{C_r} \frac{e^{i2z}}{z^2} dz}_{-\pi i \operatorname{Res} \left\{ \frac{e^{i2z}}{z^2} \right\}_{z=0}} - \underbrace{\int_{C_R} \frac{e^{i2z}}{z^2} dz}_{\substack{0 \\ \text{Jordan's} \\ \text{Lemma}}} = \pi i \operatorname{Res} \left\{ \frac{e^{i2z}}{z^2} \right\}_{z=0} = \pi i (2i) = -2\pi \end{aligned}$$

$$* \frac{e^{i2z}}{z^2} = \frac{1 + \underbrace{2i z + \dots}_{a_{-1}}}{z^2}$$

$$\Rightarrow \int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2} dx = -\frac{1}{2} \operatorname{Re} \{-2\pi\} = \pi \quad (\text{not } \pi/2 \text{ as the book states})$$

11.8.11 A quantum mechanical calculation of a transition probability leads to the function $f(t, \omega) = 2(1 - \cos \omega t)/\omega^2$. Show that

$$\int_{-\infty}^{\infty} f(t, \omega) d\omega = 2\pi t.$$

The solution to this problem is identical to Q9 but with a factor of 4 multiplied in the solution & $t=2$.

$$\text{Q9} \quad \frac{\frac{1}{2}(1 - \cos 2x)}{x^2}$$

$$\text{Q11} \quad \frac{2(1 - \cos \omega t)}{\omega^2}$$

Skipping to the last step of Q9, we have

$$\int_{-\infty}^{\infty} f(\omega, t) d\omega = 4(-\frac{1}{2}) \operatorname{Re}\{-t\pi\} = 2\pi t$$

11.8.12 Show that ($a > 0$):

$$(a) \int_{-\infty}^{\infty} \frac{\cos x}{x^2 + a^2} dx = \frac{\pi}{a} e^{-a}.$$

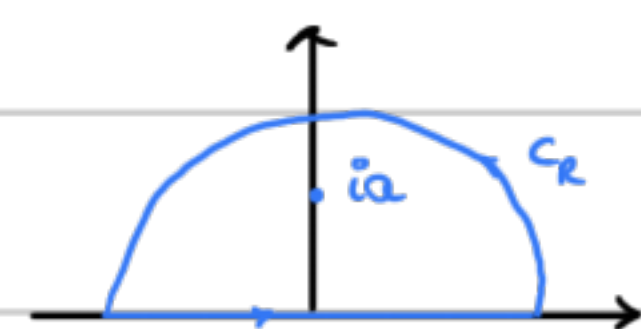
How is the right side modified if $\cos x$ is replaced by $\cos kx$?

$$(b) \int_{-\infty}^{\infty} \frac{x \sin x}{x^2 + a^2} dx = \pi e^{-a}.$$

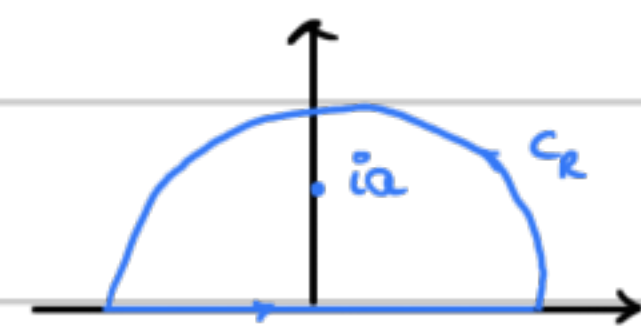
How is the right side modified if $\sin x$ is replaced by $\sin kx$?

The strategy here is to turn the trig function to e^{ix} & then use Jordan's Lemma to evaluate the semi-circle contour. We note there aren't any singularities in the range of integration on the real axis

$$\begin{aligned} a) \int_{-\infty}^{\infty} \frac{\cos x}{x^2 + a^2} dx &= \operatorname{Re} \left\{ \oint_C \frac{e^{iz}}{z^2 + a^2} dz - \int_{C_R} \frac{e^{iz}}{z^2 + a^2} dz \right\} \\ &= \operatorname{Re} \left\{ 2\pi i \operatorname{Res}_{z \rightarrow ia} \left\{ \frac{e^{iz}}{z^2 + a^2} \right\} \right\} \quad \text{0 by Jordan's Lemma} \\ &= \operatorname{Re} \left\{ 2\pi i \frac{e^{-a}}{2ia} \right\} = \frac{\pi}{a} e^{-a} \end{aligned}$$



$$\begin{aligned} b) \int_{-\infty}^{\infty} \frac{x \sin x}{x^2 + a^2} dx &= \operatorname{Im} \left\{ \oint_C \frac{ze^{iz}}{z^2 + a^2} dz - \int_{C_R} \frac{ze^{iz}}{z^2 + a^2} dz \right\} \\ &= \operatorname{Im} \left\{ 2\pi i \operatorname{Res}_{z \rightarrow ia} \left\{ \frac{ze^{iz}}{z^2 + a^2} \right\} \right\} \quad \text{0 by Jordan's Lemma} \\ &= \operatorname{Im} \left\{ 2\pi i \frac{iae^{-a}}{2ia} \right\} = \pi e^{-a} \end{aligned}$$



If $\sin x$ or $\cos x$ were replaced by $\sin kx$ or $\cos kx$, then we would just have a factor of k in the exponential in the answer

$$a) I = \frac{\pi}{a} e^{-ka}$$

$$b) I = \pi e^{-ka}$$

11.8.15 Show that $\int_0^{\infty} \frac{dx}{(x^2 + a^2)^2} = \frac{\pi}{4a^3}, \quad a > 0.$

Goal is find $\int_0^{\infty} \frac{dx}{(x^2 + a^2)^2}$

Solution

$$\int_0^{\infty} \frac{dx}{(x^2 + a^2)^2} = \frac{1}{2} \int_{-\infty}^{\infty} \frac{dx}{(x^2 + a^2)^2}$$

$$= \frac{1}{2} \oint_C \frac{dz}{(z^2 + a^2)^2} - \frac{1}{2} \int_{C_1} \frac{dz}{(z^2 + a^2)^2}$$

since $\frac{1}{(x^2 + a^2)^2}$ is even

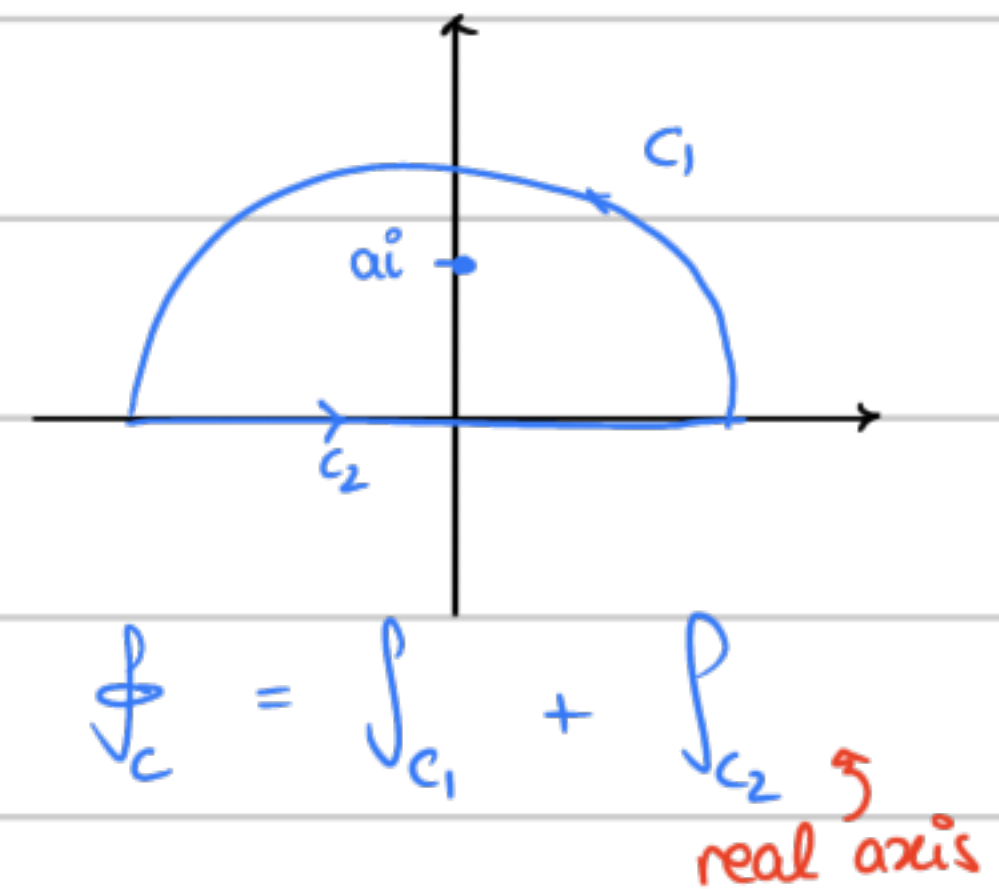
since $\frac{1}{z^4}$ decays faster than $\frac{1}{z}$

$$= \pi i \operatorname{Res}_{z=ia} \left[\frac{1}{(z^2 + a^2)^2} \right]$$

$$= \pi i \frac{-1}{4a^3} i$$

from 11.7.1 b

$$= \pi/4a^3$$

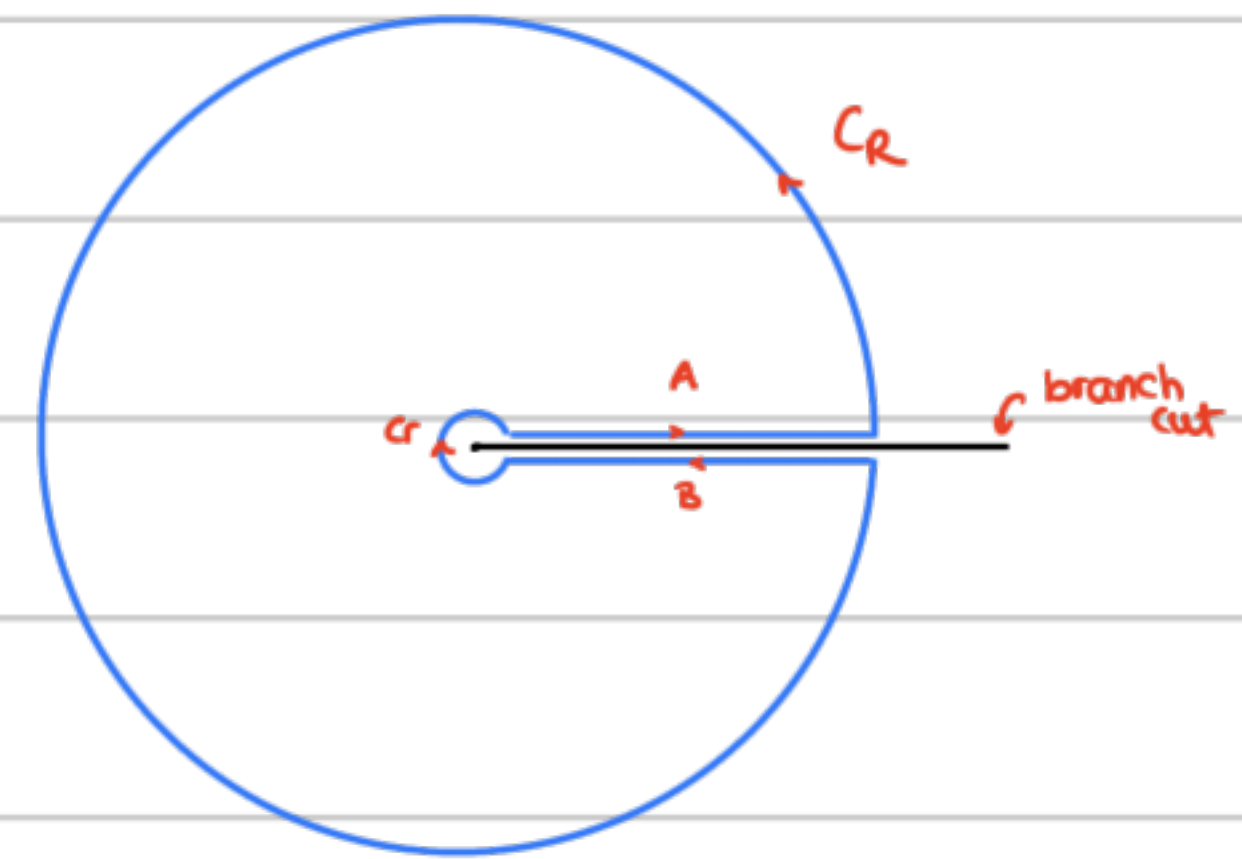


11.8.17 Evaluate $\int_0^\infty \frac{x^p \ln x}{x^2 + 1} dx, \quad 0 < p < 1.$

ANS. $\frac{\pi^2}{4} \frac{\sin(\pi p/2)}{\cos^2(\pi p/2)}.$

This looks like a place that we can exploit branch cut like Ex 11.8.8.

$$\oint_C \frac{z^p \ln z}{z^2 + 1} dz = \int_{C_R} + \int_{C_r} + \int_A + \int_B \frac{z^p \ln z}{z^2 + 1} dz$$



We evaluate all of these separately.

$$\int_{C_R} \frac{z^p \ln z}{z^2 + 1} dz = \lim_{R \rightarrow \infty} \int_0^{2\pi} \frac{R^p e^{i\theta p} (\ln R + i\theta)}{R^2 e^{i2\theta} + 1} i R e^{i\theta} d\theta = 0$$

Behavior here $\lim_{R \rightarrow \infty} O\left(\frac{R^p (\ln R) R}{R^2}\right) \sim O\left(\frac{\ln R}{R^{1-p}}\right) \quad 1-p > 0$

By L'Hopital $\lim_{R \rightarrow \infty} O\left(\frac{1-p}{R \cdot R^{-p}}\right) = 0$

$$\int_{C_r} \frac{z^p \ln z}{z^2 + 1} dz = \lim_{r \rightarrow 0} \int_0^{2\pi} \frac{r^p e^{i\theta p} (\ln r + i\theta)}{r^2 e^{i2\theta} + 1} i r e^{i\theta} d\theta = 0 \quad \text{from direct substitution of } r=0.$$

$$\int_A \frac{z^p \ln z}{z^2 + 1} dz = \int_0^\infty \frac{r^p \ln r}{r^2 + 1} dr = I \quad \text{exactly the integral we are trying to evaluate!}$$

$\nwarrow \theta=0.$

$$\int_B \frac{z^p \ln z}{z^2 + 1} dz = \int_\infty^0 \frac{r^p e^{i2\pi p} (\ln r + 2\pi i)}{r^2 e^{i4\pi} + 1} e^{i2\pi} dr \quad \sim \theta=2\pi$$

$$= - \left[\underbrace{\int_0^\infty \frac{r^p \ln r}{r^2 + 1} e^{i2\pi p} dr}_{I e^{i2\pi p}} + \underbrace{2\pi i e^{i2\pi p} \int_0^\infty \frac{r^p}{r^2 + 1} dr}_{I_1} \right]$$

We now evaluate the closed contour integral & I_1 to knock off.

$$\begin{aligned} \oint_C \frac{z^p \ln z}{z^2 + 1} dz &= 2\pi i \left[\text{Res}_{z=i} + \text{Res}_{z=-i} \right] \\ &= 2\pi i \left[\frac{\pi}{4} (e^{i\pi/2 p} - 3e^{i3\pi/2 p}) \right] \\ &= \frac{\pi^2 i}{2} [e^{i\pi/2 p} - 3e^{i3\pi/2 p}] \end{aligned}$$

We evaluate the residues on the next page

$$\text{Res}_{z=i} = \frac{z^p \ln z}{z+i} \Big|_{z=i} = \frac{i^p \ln i}{2i} = \frac{e^{i\pi/2 p} (\pi/2 i)}{2i} = -\frac{\pi}{4} e^{i\pi/2 p}$$

$$\text{Res}_{z=-i} = \frac{z^p \ln z}{z-i} \Big|_{z=-i} = \frac{(-i)^p \ln(-i)}{-2i} = \frac{e^{i3\pi/2 p} (\frac{3\pi}{2} i)}{-2i} = \frac{3\pi}{4} e^{i3\pi/2 p}$$

$$\int_0^\infty \frac{r^p}{r^2+1} dr = \frac{\pi (e^{i\pi/2 p} - e^{i3\pi/2 p})}{1 - e^{2\pi i}}$$

We put everything together now :

$$\oint_C = \int_A + \int_B$$

$$\frac{\pi^2 i}{2} [e^{i\pi/2 p} - 3e^{i3\pi/2 p}] = I - Ie^{i2\pi p} - \frac{2\pi i e^{2\pi i} \pi (e^{i\pi/2 p} - e^{i3\pi/2 p})}{1 - e^{2\pi i}}$$

$$I = \frac{\pi^2 i}{2} [e^{i\pi/2 p} - 3e^{i3\pi/2 p}] + \frac{2\pi^2 i (e^{i\pi/2 p} - e^{i3\pi/2 p}) e^{2\pi i}}{1 - e^{2\pi i}}$$

$$= \frac{\pi^2 i}{2} \frac{[(e^{i\pi/2 p} - 3e^{i3\pi/2 p}) (1 - e^{2\pi i}) + 4 (e^{i\pi/2 p} - e^{i3\pi/2 p}) e^{2\pi i}]}{(1 - e^{2\pi i})^2} \quad \begin{matrix} \times e^{-2\pi i} \\ \times e^{-2\pi i} \end{matrix}$$

$$= \frac{\pi^2 i}{2} \frac{[e^{i\pi/2 p} - 3e^{i3\pi/2 p} - e^{i5\pi/2 p} + 3e^{i7\pi/2 p}] e^{-2\pi i} + 4 (e^{i\pi/2 p} - e^{i3\pi/2 p}) }{-4 \sin^2 \pi p}$$

$$= \frac{\pi^2 i}{2} \frac{[e^{-i3\pi/2 p} - 3e^{-i\pi/2 p} + 3e^{i\pi/2 p} - e^{i3\pi/2 p}]}{-4 \sin^2 \pi p}$$

$$= -\frac{\pi^2 i}{8} \frac{(e^{-i\pi/2 p} - e^{i\pi/2 p})^3}{\sin^2 \pi p}$$

$$= \pi^2 \frac{\sin^3 \frac{\pi p}{2}}{\sin^2 \pi p} = \pi^2 \frac{\sin^3 \frac{\pi p}{2}}{4 \sin^2 \frac{\pi p}{2} \cos^2 \frac{\pi p}{2}} = \frac{\pi^2}{4} \frac{\sin \frac{\pi p}{2}}{\cos^2 \frac{\pi p}{2}}$$

11.8.20 Show that

$$\int_0^{\infty} \frac{x^a}{(x+1)^2} dx = \frac{\pi a}{\sin \pi a},$$

where $-1 < a < 1$.

Hint. Use the contour shown in Fig. 11.26, noting that $z = 0$ is a branch point and the positive x -axis can be chosen to be a cut line.

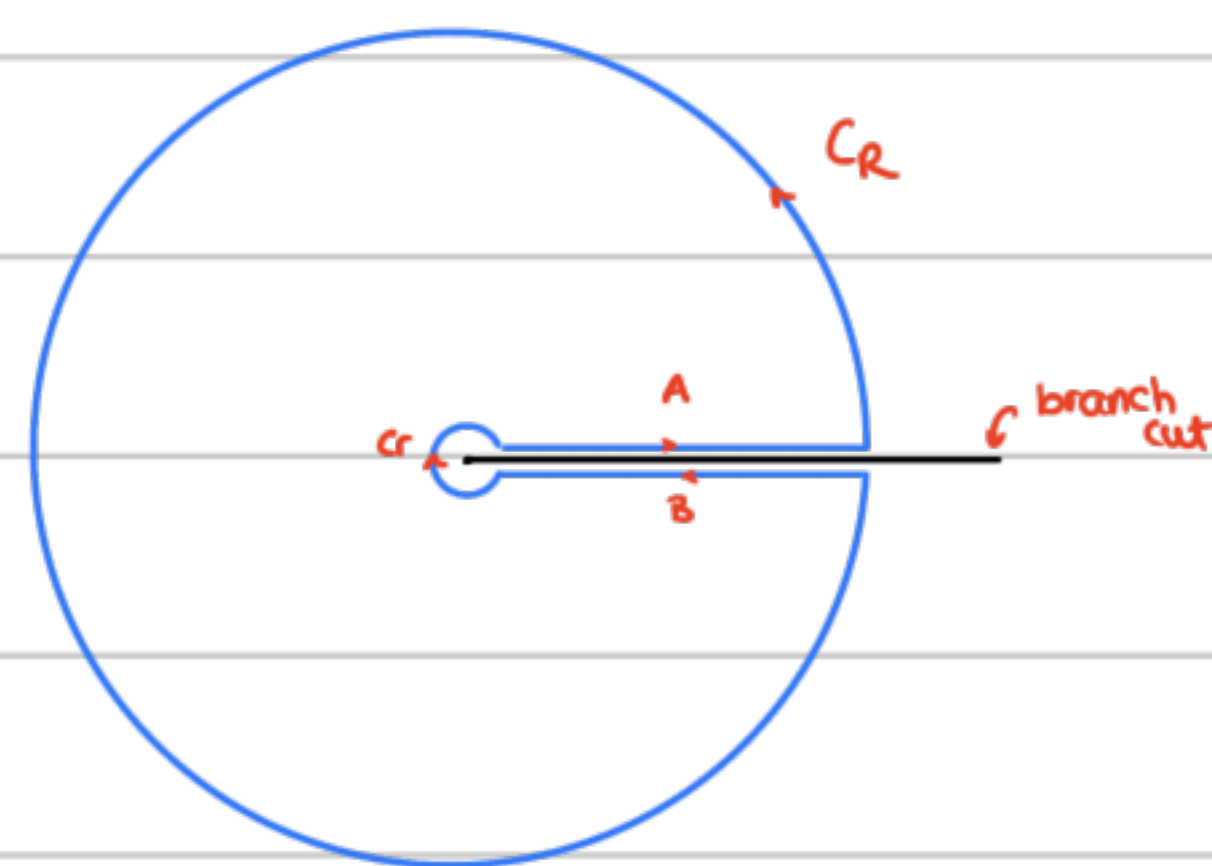
We see a branch point from the x^a term that we may be able to exploit!

$$\oint_C \frac{z^a}{(z+1)^2} dz = \int_{C_R} + \int_{C_r} + \int_A + \int_B \frac{z^a}{(z+1)^2} dz \quad (1)$$

We again evaluate each integral individually &

$$\oint_C \frac{z^a}{(z+1)^2} dz = 2\pi i \operatorname{Res}_{z=-1} \left[\frac{z^a}{(z+1)^2} \right]$$

$$= 2\pi i \lim_{z \rightarrow -1} \frac{d}{dz} z^a = 2\pi i \lim_{z \rightarrow -1} a z^{a-1} = 2\pi i a (-1)^{a-1} \quad \text{or} \quad 2\pi i a e^{i\pi(a-1)}$$



$$\int_{C_R} \frac{z^a}{(z+1)^2} dz \rightarrow 0 \quad \text{since } f(z) \sim O(z^{a-2}) \quad \text{which decays faster than } 1/z$$

$$\int_{C_r} \frac{z^a}{(z+1)^2} dz \rightarrow 0 \quad \text{since no singularities at } z=0$$

$$\int_A \frac{z^a}{(z+1)^2} dz = \int_0^{\infty} \frac{r^a}{(r+1)^2} dr = I \quad \begin{array}{l} \text{Integral} \\ \text{we are trying to find!} \end{array}$$

$$\int_B \frac{z^a}{(z+1)^2} dz = \int_{\infty}^0 \frac{r^a e^{i2\pi a}}{(re^{i2\pi} + 1)^2} e^{i2\pi} dr = -e^{i2\pi a} I$$

Plugging all the integral values in (1)

$$2\pi i a e^{i\pi(a-1)} = I(1 - e^{i2\pi a})$$

$$I = \frac{2\pi i a e^{i\pi(a-1)}}{1 - e^{i2\pi a}} = \frac{2\pi i a e^{-i\pi}}{-2i \sin \pi a} = \frac{\pi a}{\sin \pi a}$$

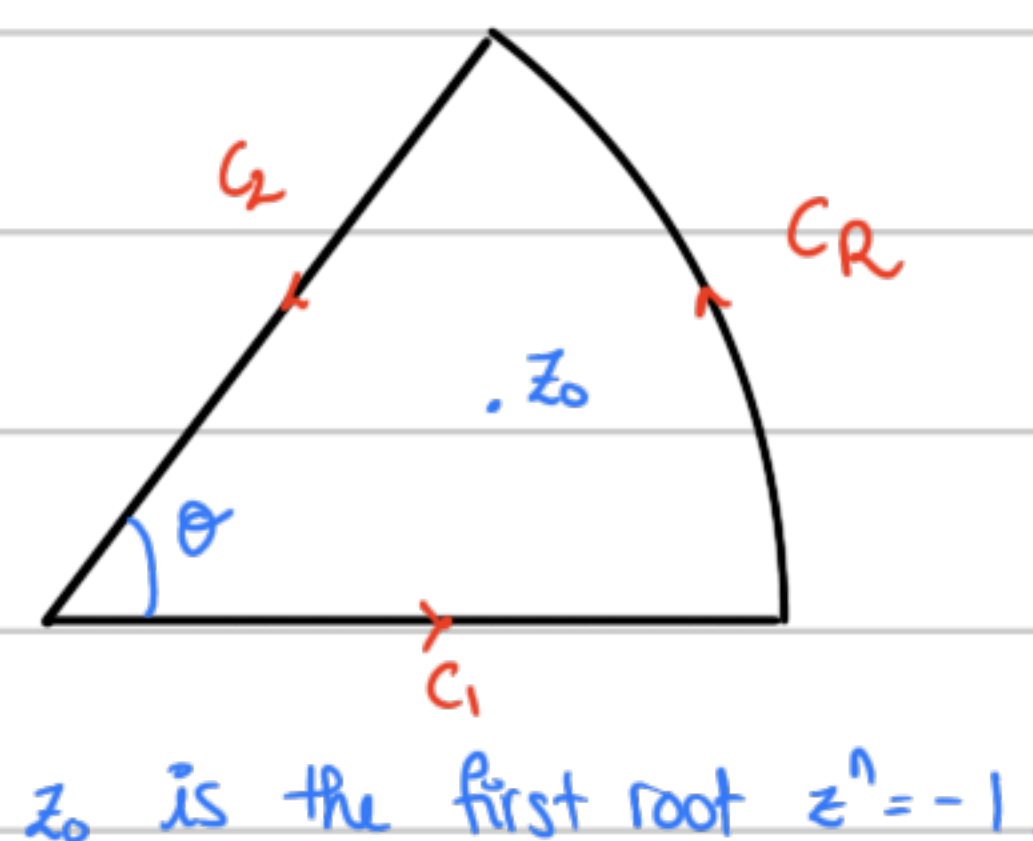
11.8.22 Show that

$$\int_0^{\infty} \frac{dx}{1+x^n} = \frac{\pi/n}{\sin(\pi/n)}.$$

Hint. Try the contour shown in Fig. 11.30, with $\theta = 2\pi/n$.

Proof of the value of this integration is near identical to Example 11.8.6 for $\int \frac{dx}{1+x^3}$

We attempt to evaluate the integral on a sector since $\frac{1}{1+x^n}$ need not necessarily be even making evaluating on a semi-circle difficult. However, we will be able to relate C_1 & C_2 with a careful choice of θ .



$$\oint_C \frac{dz}{1+z^n} = \int_{C_1} + \int_{C_2} + \int_{C_3} \frac{dz}{1+z^n} \quad (1)$$

We again evaluate each integral individually &

$$\begin{aligned} \oint_C \frac{dz}{1+z^n} &= 2\pi i \operatorname{Res}_{z=z_0} \frac{1}{1+z^n} \\ &= 2\pi i \lim_{z \rightarrow z_0} \frac{z-z_0}{1+z^n} \end{aligned}$$

The contour of C will always include one root of $z^n = -1$ & that is $z_0 = (-1)^{1/n} = e^{i\pi/n}$

$$= \frac{2\pi i}{n z^{n-1}} \Big|_{z=e^{i\pi/n}} = \frac{2\pi i}{n e^{i\pi(n-1)/n}}$$

$$\int_{C_3} \frac{dz}{1+z^n} \rightarrow 0 \quad \text{since } \frac{1}{z^n} \text{ decays faster than } \frac{1}{z}$$

$$\int_{C_1} \frac{dz}{1+z^n} = \int_0^{\infty} \frac{dr}{1+r^n} = I \quad \text{Integral we are trying to find!}$$

$$\int_{C_2} \frac{dz}{1+z^n} = \int_0^{\infty} \frac{e^{i2\pi/n} dr}{1+r^n e^{i2\pi/n}} = -e^{i2\pi/n} \int_0^{\infty} \frac{dr}{1+r^n} = -e^{i2\pi/n} I$$

Plugging all the integral values in (1)

$$\frac{2\pi i}{n e^{i\pi(n-1)/n}} = I(1 - e^{i2\pi/n})$$

$$I = \frac{2\pi i}{n e^{i\pi(n-1)/n}} \cdot \frac{1}{1 - e^{i2\pi/n}} \times \frac{e^{-i\pi/n}}{e^{-i\pi/n}}$$

$$= \frac{2\pi i}{n e^{i\pi}} \cdot \frac{1}{-2i \sin \pi/n} = \frac{\pi/n}{\sin \pi/n}$$

11.8.24 Show that

$$\int_0^{\infty} \frac{x^{-a}}{x+1} dx = \frac{\pi}{\sin a\pi},$$

where $0 < a < 1$.

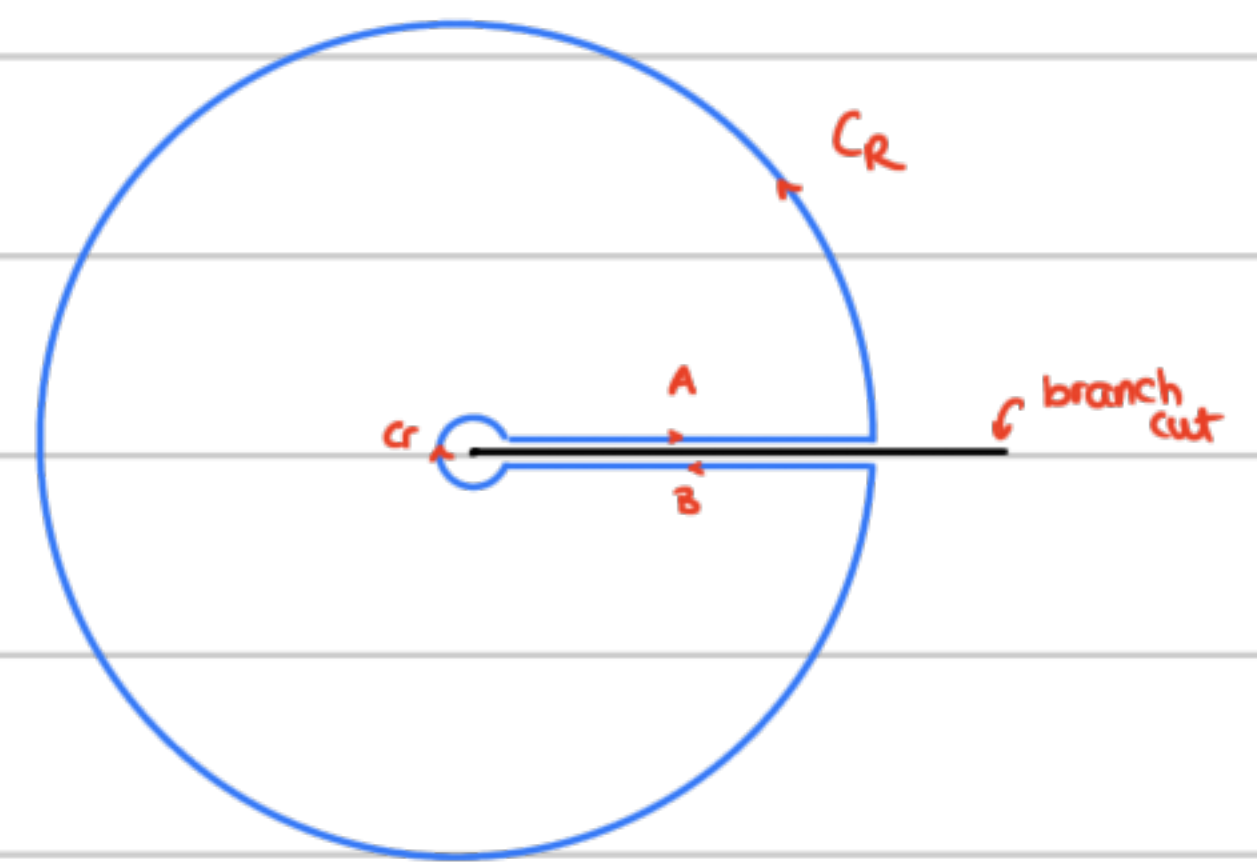
Hint. You have a branch point and you will need a cut line. Try the contour shown in Fig. 11.26.

This is almost identical to 11.8.20 but with an $x+1$ instead of $(x+1)^2$ in the denominator

$$\oint_C \frac{z^{-a}}{z+1} dz = \int_{C_R} + \int_{C_r} + \int_A + \int_B \frac{z^{-a}}{z+1} dz \quad (1)$$

We again evaluate each integral individually &

$$\begin{aligned} \oint_C \frac{z^{-a}}{z+1} dz &= 2\pi i \operatorname{Res}_{z=-1} \left[\frac{z^{-a}}{z+1} \right] \\ &= 2\pi i (-1)^{-a} \quad \text{or} \quad 2\pi i e^{-ia} \end{aligned}$$



$$\int_{C_R} \frac{z^{-a}}{z+1} dz \rightarrow 0 \quad \text{since } f(z) \sim O(z^{-(a+1)}) \quad \text{which decays faster than } 1/z$$

$$\int_{C_r} \frac{z^{-a}}{z+1} dz \rightarrow 0 \quad \text{since no singularities at } z=0$$

$$\int_A \frac{z^{-a}}{z+1} dz = \int_0^{\infty} \frac{r^{-a}}{r+1} dr = I \quad \begin{array}{l} \text{Integral} \\ \swarrow \text{we are trying to find!} \end{array}$$

$$\int_B \frac{z^{-a}}{z+1} dz = \int_{\infty}^0 \frac{r^{-a} e^{-i2\pi a}}{r e^{i2\pi} + 1} e^{i2\pi} dr = -e^{-i2\pi a} I$$

Putting all the integral values in (1)

$$2\pi i e^{-ia} = I (1 - e^{-i2\pi a})$$

$$I = \frac{2\pi i e^{-ia}}{1 - e^{-i2\pi a}} \quad \begin{array}{l} \times e^{ia} \\ \times e^{ia} \end{array}$$

$$= \frac{2\pi i}{2i \sin \pi a} = \frac{\pi}{\sin \pi a}.$$